LAB SHEET

K.R. MANGALAM UNIVERSITY

SOET

(School of Engineering & Technology)

B.TECH CSE (CORE)

Database & Management System Lab (ENCS254)

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LAB SEET – 2

Objective: To understand and implement SQL transactions using COMMIT, SAVEPOINT, and ROLLBACK, retrieve records using SELECT queries, and perform INSERT, UPDATE, and ALTER operations on tables.

- **DDL Questions (Data Definition Language)**

1. Create table CLASS

```sql
CREATE DATABASE LAB_SHEET_2;
USE LAB_SHEET_2;
CREATE TABLE CLASS (
    ID INT PRIMARY KEY,
    Name VARCHAR(20)
);
INSERT INTO CLASS VALUES (1,'Bravo');
INSERT INTO CLASS VALUES (2,'Alex');
INSERT INTO CLASS VALUES (4,'Cheng');
```

2. Create table EMPLOYEES

```sql
CREATE TABLE EMPLOYEES (
    Employee_Id INT PRIMARY KEY,
    First_Name VARCHAR(50),
    Last_Name VARCHAR(50),
    Email VARCHAR(100),
    Phone_Number VARCHAR(20),
    Hire_Date DATE,
    Job_Id VARCHAR(20),
    Salary INT,
    Commission_Pct INT,
    Manager_Id INT,
    Department_Id INT
);
INSERT INTO EMPLOYEES VALUES
(101,'Neha','Sharma','neha@gmail.com','9876543210','2023-01-10','HR',5000,NULL,100,60);
INSERT INTO EMPLOYEES VALUES
(102,'Rohan','AUSTIN','rohan@gmail.com','9876543211','2022-03-15','IT',7000,NULL,100,70);
INSERT INTO EMPLOYEES VALUES
(103,'Aman','Verma','aman@gmail.com','9876543212','2021-07-20','Sales',4500,NULL,101,80);
INSERT INTO EMPLOYEES VALUES
(104,'Priya','Kapoor','priya@gmail.com','9876543213','2020-11-25','Finance',4800,NULL,102,60);
INSERT INTO EMPLOYEES VALUES
(105,'Karan','Mehta','karan@gmail.com','9876543214','2019-05-30','IT',5500,NULL,100,80);
```

3. Create table Client_master

```
CREATE TABLE Client_master (
    ClientNO VARCHAR(10) PRIMARY KEY,
    Name VARCHAR(50),
    Address VARCHAR(100),
    City VARCHAR(50),
    Pincode INT,
    State VARCHAR(50),
    Bal_due INT
);
```

4. Rename table Client_master to Client12

```
RENAME TABLE Client_master TO Client12;
```

5. Display bal_due column heading as “BALANCE”

```
SELECT Bal_due AS "BALANCE"
FROM Client_master;
```

- **DML Questions (Data Manipulation Language)**

➢ **Part A: Transactions and Savepoints (CLASS Table)**

1. COMMIT

```
INSERT INTO CLASS VALUES (5,'Rahul');
COMMIT;
```

2. SAVEPOINT A

```
UPDATE CLASS SET Name = 'Abhijeet' WHERE ID = 5;
SAVEPOINT A;
```

3. SAVEPOINT B

```
INSERT INTO CLASS VALUES (6,'Chris');
SAVEPOINT B;
```

4. SAVEPOINT C

```
INSERT INTO CLASS VALUES (7,'Bravo');
SAVEPOINT C;
```

5. SELECT ALL FROM CLASS

```
SELECT * FROM CLASS;
```

Output:

Result Grid

ID	Name
1	Bravo
2	Alex
4	Cheng
5	Abhijeet
6	Chris
7	Bravo
NULL	NULL

6. ROLLBACK TO B AND SELECT ALL FROM CLASS

```
ROLLBACK TO B;
SELECT * FROM CLASS;
```

Output:

Result Grid

ID	Name
1	Bravo
2	Alex
4	Cheng
5	Abhijeet
6	Chris
NULL	NULL

8. ROLLBACK TO A AND SELECT ALL FROM CLASS

```
ROLLBACK TO A;
SELECT * FROM CLASS;
```

Output:

Result Grid

ID	Name
1	Bravo
2	Alex
4	Cheng
5	Abhijeet
NULL	NULL

➢ Part B: Retrieving Records (EMPLOYEES Table)

1. Employee id, names and salaries

```
SELECT Employee_Id, First_Name, Last_Name, Salary
FROM EMPLOYEES;
```

OUTPUT:

Result Grid | Filter Rows: | Ed

Employee_Id	First_Name	Last_Name	Salary
101	Neha	Sharma	5000
102	Rohan	AUSTIN	7000
103	Aman	Verma	4500
104	Priya	Kapoor	4800
105	Karan	Mehta	5500
NULL	NULL	NULL	NULL

2. Employees under manager 100

```
SELECT *
FROM EMPLOYEES
WHERE Manager_Id = 100;
```

Output:

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

Employee_Id	First_Name	Last_Name	Email	Phone_Number	Hire_Date	Job_Id	Salary	Commission_Pct	Manager_Id	Department_Id
101	Neha	Sharma	neha@gmail.com	9876543210	2023-01-10	HR	5000	NULL	100	60
102	Rohan	AUSTIN	rohan@gmail.com	9876543211	2022-03-15	IT	7000	NULL	100	70
105	Karan	Mehta	karan@gmail.com	9876543214	2019-05-30	IT	5500	NULL	100	80
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

3. Salary ≥ 4800

```
SELECT First_Name, Last_Name
FROM EMPLOYEES
WHERE Salary >= 4800;
```

Output:

Result Grid | Filter Rows:

First_Name	Last_Name
Neha	Sharma
Rohan	AUSTIN
Priya	Kapoor
Karan	Mehta

4. Last name = 'AUSTIN'

```
SELECT *
FROM EMPLOYEES
WHERE Last_Name = 'AUSTIN';
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

Employee_Id	First_Name	Last_Name	Email	Phone_Number	Hire_Date	Job_Id	Salary	Commission_Pct	Manager_Id	Department_Id
102	Rohan	AUSTIN	rohan@gmail.com	9876543211	2022-03-15	IT	7000	NULL	100	70
NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL	NULL

5. Department 60, 70, 80

```
SELECT First_Name, Last_Name
FROM EMPLOYEES
WHERE Department_Id IN (60,70,80);
```

Output:

Result Grid | Filter Rows:

First_Name	Last_Name
Neha	Sharma
Rohan	AUSTIN
Aman	Verma
Priya	Kapoor
Karan	Mehta

6. Display unique Manager_Id

```
SELECT DISTINCT Manager_Id
FROM EMPLOYEES;
```

Output:

Result Grid |

Manager_Id
100
101
102

➢ Part C: Updating Records (Client_master Table)

1. Insert 5 records

```
INSERT INTO Client12 VALUES
('C123','Rohan','Delhi','Delhi',110001,'Delhi',6000);

INSERT INTO Client12 VALUES
('C124','Priya','Mumbai','Mumbai',400001,'Maharashtra',3000);

INSERT INTO Client12 VALUES
('C125','Aman','Jaipur','Jaipur',302001,'Rajasthan',8000);

INSERT INTO Client12 VALUES
('C126','Neha','Lucknow','Lucknow',226001,'UP',2000);

INSERT INTO Client12 VALUES
('C127','Karan','Chandigarh','Chandigarh',160001,'Punjab',9000);
```

2. Clients with bal_due > 5000

```
SELECT Name
FROM Client12
WHERE Bal_due > 5000;
```

Output:

Result Grid

Name
Rohan
Aman
Karan

3. Change bal_due of ClientNO "C123" to 5100

```
UPDATE Client12
SET Bal_due = 5100
WHERE ClientNO = 'C123';
```

Output:

Result Grid | Filter R

ClientNO	Bal_due
C123	5100
NULL	NULL